

KTern.AI

Product Requirements Document

Agentic AI Driven License Optimization & SoD Analysis

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Any enhancement outside the Statement of Work (SoW) will be treated as a separate enhancement ticket and governed under the KTern.AI SLA Policy. Such items are clearly marked throughout this document.

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1. Document Overview

This Product Requirements Document (PRD) defines the functional requirements, expected system behavior, and feature scope for KTern.AI's Agentic AI-driven License Optimization and Segregation of Duties (SoD) Analysis modules, as contracted with Lotte Chemical.

The document is structured in two primary streams:

- **Stream 1:** License Optimization — AI-driven analysis of SAP user licenses, roles, transactions, and authorization objects to identify optimization opportunities and reduce costs.
 - The proposed pricing is applicable for 12 runs per year for a duration of 3 years.
- **Stream 2:** SoD (Segregation of Duties) Analysis — AI-powered detection of access conflicts, critical violations, and compliance gaps across SAP authorization landscapes.
 - The proposed pricing is applicable for 90 runs per year for a duration of 3 years.

Each feature in this document is tagged as either:

✓ **IN SCOPE (SoW)** Covered under the signed Statement of Work — delivered as part of the KTern.AI subscription.

⚠ **Enhancement Request Ticket Required**

[This requires a separate ticket processed under the KTern.AI SLA]

2. Need Statement

Lotte Chemical manages a large-scale, multi-geography SAP landscape. Maintaining SAP license compliance, cost efficiency, and access governance at this scale using manual methods is no longer sustainable. The key pain points driving this initiative include:

- No single source of truth for SAP user license consumption across business units and geographies.
- Manual license classification requiring significant effort from SAP BASIS and Security teams.
- Lack of real-time visibility into role usage, authorization footprints, and SoD risk exposure.
- Inability to proactively detect compliance violations prior to audits.
- High cost of over-licensed users who do not consume the access they are assigned.
- No repeatable, scalable governance model for ongoing license and SoD management.

KTern.AI addresses these pain points through its Agentic AI-driven License Optimization and SoD Analysis agents, delivering automated, continuous, and audit-ready intelligence across the SAP landscape.

3. Scope of the Project

3.1 In-Scope (Statement of Work)

Stream	Capability Area	Details
License Optimization	User Information Analysis	Automated ingestion of user master data, validity , status, license assignments
License Optimization	License Analysis & Classification	Mapping licenses to actual usage; identifying mismatches and over-licensed users [Recommendations]
License Optimization	Usage Statistics	Login frequency, transaction execution trends, transaction dormancy detection, Trace should be turned on in the SAP System (Automated ingestion without manual data extraction from SAP)
License Optimization	Assigned Roles Analysis	Redundancy, overlap, and relevance of composite and single roles - Recommendations
License Optimization	Authorization Objects Overview	Consolidated view of all auth objects with frequency of use
License Optimization	Assigned Transactions Analysis	T-Code mapping with execution history; unused transaction detection, Last Login Date and Login Frequency
License Optimization	License Breakdown	Granular breakdown of license types per user mapped to actual usage
License Optimization	Role Utilization Analysis	Unused + used, and redundant roles at scale
License Optimization	Role Activity & Auth Analysis	Object-level, field-level, and activity-level role decoding
License Optimization	Dashboard & Export	Home Page dashboard with analytics; downloadable Excel/PDF reports
License Optimization	Dashboard	Target License Classification column with hierarchy logic (Productivity < Functional < Professional) – this we were showing earlier inside each users- Should be shown both inside and outside the home screen
SoD Analysis	Executive Summary	AI-generated compliance summary with risk scoring

SoD Analysis	Overall Compliance Assessment	Compliance posture benchmarked against SAP GRC best practices
SoD Analysis	Critical Findings Summary	High-risk violation consolidation with severity scoring
SoD Analysis	Immediate Actions (24–48 hrs)	Urgent remediation recommendations with step-by-step guidance
SoD Analysis	Super Administrators Detection	Unrestricted access detection across SAP systems
SoD Analysis	Dual Process Control Violations	Cross-role, cross-transaction SoD conflict identification
SoD Analysis	Emergency Access Analysis	Firefighter access pattern detection and misuse flags
SoD Analysis	OTC Control Violations	End-to-end Order-to-Cash control detection
SoD Analysis	High-Risk Service Accounts	Unmanaged service and technical account privilege analysis
SoD Analysis	Remediation Recommendations	AI-backed role redesign, access removal, and governance policies
SoD Analysis	Continuous Compliance Enablement	Ongoing SoD monitoring via scans and dashboards

3.2 Enhancement Tickets

The following enhancements are outside the signed SoW and will be treated as enhancement tickets processed under the KTern.AI SLA. They are detailed in Sections 5 and 6 below.

1. User-Based License Consumption card on Home Page dashboard

Card1: User-Based License Consumption (need in landing page and home page)

- **Columns:**
- User Name
- License Count by Type (Tree View)
- Professional (count)
- Functional (count)
- Productive / Core (count)
- Target License Classification

- Expected Structure:
 - User Name
 - └ License Type
 - └ No. of Professional Licenses
 - └ No. of Functional Licenses
 - └ No. of Productive Licenses
 - └ Target License Classification
- Behaviour:
 - Expand user → view license types and counts [Have expand all - collapse all option]
 - Auto-Drill-down → list contributing roles / objects
 - Highlight **final target classification per user**
 - Ensure to have download icon & filters
 - Sample but few of the expectations mentioned is not below [other extract columns not mentioned aren't required]:

Selection: All / Total (166)

User	Full Name	ID	Current Classification	Target Classification	Ratio	Ref.Us.	User Group	Last Logon	Count
<u>ELLIOTT</u>	Elliott Allison	GB	S/4HANA Cloud advanced use	GB Advanced Use	9/56		MANAGEMENT	04/16/2026	1
<u>AGUIDRY</u>	Amadia Guidry	HC	S/4HANA EM Functional Use	GB Advanced Use	2/31			04/09/2026	1
<u>ALLA GREEN</u>	Alla Green		Not Classified (Default User Type)	GB Advanced Use	10/60		MANAGEMENT	04/16/2026	1
<u>AMANDA KOLB</u>	Amanda Kolb		Not Classified (Default User Type)	GB Advanced Use	8/52			04/16/2026	1
<u>APOCHE</u>	Adaam Poche		Not Classified (Default User Type)	GB Advanced Use	3/47			04/10/2026	1
<u>ASCHEXNIDER</u>	Andre Schexnider		Not Classified (Default User Type)	GB Advanced Use	1/30			04/17/2026	1
<u>ASHTON BACON</u>	Ashton Bacon		Not Classified (Default User Type)	GB Advanced Use	1/31			04/16/2026	1
<u>AVINGENT</u>	Austin Vincent		Not Classified (Default User Type)	GC Core Use	1/16			03/31/2026	1
<u>BBREVES</u>	Bruno Breves	HC	S/4HANA EM Functional Use	GB Advanced Use	21/100		MANAGEMENT	04/16/2026	1
<u>BBURNS</u>	Ben Burns		Not Classified (Default User Type)	GB Advanced Use	2/21		MANAGEMENT	04/17/2026	1
<u>BCARRIER</u>	Baron Carrier	HC	S/4HANA EM Functional Use	GB Advanced Use	5/48			04/17/2026	1
<u>BDEROUEN</u>	Bryon Derouen		Not Classified (Default User Type)	GB Advanced Use	2/40			02/18/2026	1
<u>BFERCHAUD</u>	Brittney Ferchaud		Not Classified (Default User Type)	GB Advanced Use	11/55			04/16/2026	1
<u>BOILL</u>	Brooklyn Gill		Not Classified (Default User Type)	GB Advanced Use	8/52			04/16/2026	1
<u>BHOOPER</u>	Brett Hooper		Not Classified (Default User Type)	GB Advanced Use	3/42			04/01/2026	1
<u>BJANI</u>	Bhavik Jani		Not Classified (Default User Type)	GB Advanced Use	3/43			04/14/2026	1
<u>BKERSHAW</u>	Ben Kershaw		Not Classified (Default User Type)	GB Advanced Use	1/15			04/16/2026	1
<u>BLEBLANC</u>	Barrett Leblanc		Not Classified (Default User Type)	GC Core Use	6/39			01/06/2026	1
<u>BQUEBODEAUX</u>	BAY QUEBODEAUX		Not Classified (Default User Type)	GC Core Use	6/26			04/15/2026	1
<u>BRENTROP</u>	Bruce Rentrop		Not Classified (Default User Type)	GC Core Use	1/16			04/09/2026	1
<u>BRIAN KWAK</u>	Brian Kwak		Not Classified (Default User Type)	GB Advanced Use	3/31			04/17/2026	1
<u>CANGE</u>	CHRISTOPHER ANGE		Not Classified (Default User Type)	GB Advanced Use	2/29			04/09/2026	1
<u>CARLEY SIMON</u>	Carley Simon		Not Classified (Default User Type)	GD Self-Service Use	51/51			04/13/2026	1
<u>CBRESCHER</u>	Curtis Brescher		Not Classified (Default User Type)	GB Advanced Use	1/5		MANAGEMENT	03/30/2026	1
<u>CCRAIN</u>	Chad Crain		Not Classified (Default User Type)	GD Self-Service Use	17/18			04/14/2026	1
<u>CHAD POE</u>	Chad Poe		Not Classified (Default User Type)	GC Core Use	6/43			02/25/2026	1
<u>CHLAUGHLIN</u>	Christopher Laughlin		Not Classified (Default User Type)	GC Core Use	3/13				1
<u>CHRIS GANDY</u>	CHRIS GANDY		Not Classified (Default User Type)	GC Core Use	1/16			04/06/2026	1

2. Role-Based License Classification card on Home Page dashboard

Card2: Role-Based License Classification (need in landing page and home page)

- Columns:
 - Role Name
 - Role Description

- Target License Classification
- **Behaviour:**
 - Showcase the role and have the role description
 - Indicate the "target license classification" within each role
- Ensure to have download icon & filters
- Expectations [other extract columns not mentioned aren't required]:

Role / Profile	Description	Current Classification	Target Classification	Ratio	Obj.	Cou.	Use
ZIT_BR_ALL_EMPLOYEES	Common Role For All Employees		GD Self-Service Use	109/109	593	1	107
ZPM_BR_NOTIF_CREATE_1720	Role for Creating Notification - Comapny Code 1720		GC Core Use	1/90	712	1	104
ZPM_BR_NOTIF_CREATE_1730	Role for Creating Notification - Comapny Code 1730		GC Core Use	1/90	712	1	101
ZMM_BR_PR_DISPLAY_PRO	Role for Purchase Requisition Display		GD Self-Service Use	43/43	107	1	96
ZMM_BR_PO_DISPLAY_PRO	Role for Purchase Order Display - PRO		GD Self-Service Use	61/61	231	1	96
ZMM_BR_PR_CREATE_PRO	Role for Purchase Requisition Create		GD Self-Service Use	59/59	133	1	96
ZMM_BR_PR_CHANGE_PRO	Role for Purchase Requisition Change		GD Self-Service Use	47/47	101	1	96
ZMM_BR_MNGR_APPRVL AUTH_1710	Derived role for approval auth		GB Advanced Use	3/155	442	1	76
ZPM_BR_REPORTING_1720	Role for PM reports - Company Code 1720		GD Self-Service Use	46/46	407	1	65
ZL_BR_PROM_AUTH	Role Prometheus authorizations		GD Self-Service Use	1/1	25	1	59
ZPM_BR_REPORTING_1730	Role for PM reports - Company Code 1730		GD Self-Service Use	46/46	407	1	57
ZPM_BR_PROCUREMENT_1720	Role for PM Procurement - Company Code 1720		GD Self-Service Use	186/186	513	1	54
ZPM_BR_MORD_PROCESS_1720	Role for Processing Maintenance Order - Company Code 1720		GC Core Use	1/136	940	1	53
ZPM_BR_PROCUREMENT_1730	Role for PM Procurement - Company Code 1730		GD Self-Service Use	186/186	513	1	52
ZPM_BR_MORD_PROCESS_1710	Role for Processing Maintenance Order - Company Code 1710		GC Core Use	1/136	940	1	51
ZPM_BR_MORD_PROCESS_1730	Role for Processing Maintenance Order - Company Code 1730		GC Core Use	1/136	940	1	51
ZPM_BR_PROCUREMENT_1710	Role for PM Procurement - Company Code 1710		GD Self-Service Use	186/186	513	1	51
ZPM_BR_REPORTING_1710	Role for PM reports - Company Code 1710		GD Self-Service Use	46/46	407	1	50
ZL_BR_CA_ALL_USER_1720	Role for All User - 1720		GD Self-Service Use	89/89	267	1	49
ZPM_BR_USR_MTR_DATA_MNT_1720	Derived 1720 - Role for Master Data for PM user		GD Self-Service Use	65/65	515	1	49
ZPM_BR_MORD_PLAN_1720	Role for Planning Maintenance Order - Comapny Code 1720		GD Self-Service Use	66/66	320	1	49
ZCA_BR_MM_ROLE_1720	Common apps role for all users - 1720		GB Advanced Use	2/173	748	1	49
ZPM_BR_NOTIF_CREATE_1710	Role for Creating Notification - Comapny Code 1710		GC Core Use	1/90	712	1	48
ZPM_BR_MORD_PLAN_1710	Role for Planning Maintenance Order - Comapny Code 1710		GD Self-Service Use	66/66	320	1	47
ZEWM_BR_WH_DISPLAY_MSUP_LAMW	Role for EWM - Warehouse Display - 1720 & 1730 - Maintenance Supervisor - LAMW		GD Self-Service Use	140/140	719	1	47
ZPM_BR_MORD_PLAN_1730	Role for Planning Maintenance Order - Comapny Code 1730		GD Self-Service Use	66/66	320	1	47
ZL_BR_CA_BP_OIS	Role for Display Business Partner		GD Self-Service Use	55/55	140	1	47
ZPM_BR_USR_MTR_DATA_MNT_1710	Derived 1710 - Role for Master Data for PM user		GD Self-Service Use	65/65	515	1	47

3. Recommendations – Object Level Cleaning & Role Based Cleaning

Object-Level Cleaning (Focus: Professional License Objects)

- Eg: Filter “Professional License” objects only
 - Step 1: Object Traceability: Identify all roles containing the auth object (e.g., “ZX”) - (irrespective of used/unused)
 - Step 2: Role Assignment: For each role (e.g., F_LFA1) that contains the object:
 - State the number of users assigned & List the user names
 - Step 3: Usage & License Segregation: Based on trace:
 - Identify users who actually use the object
 - Classify users into:

- High privilege (Professional required)
 - Lower privilege (Functional/Productive sufficient)
 - Step 4: Role Optimization:
 - Create a separate high-privilege role for the identified Professional users
 - Remove high-privilege objects from the old role (F_LFA1)
 - Retain only required objects for remaining users in the old role (F_LFA1)
2. **Role-Based Cleaning (Focus: Usage Efficiency)**
3. Step 1: Role Assignment Analysis: Identify all users assigned to the role
4. Step 2: Usage Validation: Identify users who:
- Did not use any objects in the role
 - Or have very low usage (specify it clearly)
5. Step 3: Pattern Identification: Evaluate across users:
- If majority unused (give count) → role is a cleanup candidate
 - If partially used (give count) → role requires optimization (object cleanup)
6. Step 4: User-Level Cleanup: For users not using the role:
- Identify other unused roles assigned
 - Recommend removal of unused role assignments

4. Key Stakeholders

Stakeholder / Role	Responsibility	Stream
SAP License Manager (Lotte Chemical)	Primary consumer of License Optimization dashboard; drives optimization decisions	License Optimization
SAP Basis / Security Team	Data provisioning, role management, access trace activation	Both
IT Compliance & Audit Team	Review SoD findings, validate remediation actions, audit readiness	SoD Analysis
Finance & Risk Leadership	Approve remediation roadmap; quantify compliance risk exposure	SoD Analysis
CIO / IT Director	Strategic oversight; license cost optimization and compliance governance	Both

KTern.AI Customer Success	Onboarding, setup, enablement, ticket management, SLA governance	Both
KTern.AI Product Engineering	Feature delivery, enhancement ticket resolution, SLA adherence	Both

5. Enhancement Ticket & SLA Policy

Any feature or enhancement that falls outside the boundaries of the signed Statement of Work shall be submitted as an enhancement ticket and processed in accordance with the KTern.AI Service Level Agreement (SLA). The SLA governs all support and enhancement activities.

Enhancement Ticket Process:

- Customer raises an enhancement request via KTern.AI support portal or Customer Success team.
- KTern.AI Product team reviews and classifies the ticket by priority (P1–P4).
- Ticket is scheduled into the monthly product release cycle based on priority and common customer demand.
- Customer is notified of the target release sprint and estimated delivery date.
- Delivery is governed by the SLA timelines detailed below.

Priority	Response Time	Resolution Target	Example
Critical (P1)	4 Business Hours	3 Business Day	Bug Fixes
High (P2)	8 Business Hours	5 Business Days	Minor Enhancement
Medium (P3)	1 Business Day	12 Business Days	Enhancement ticket
Low (P4)	1 Business Days	14 Business Days	Any UI change, cosmetic, feature request

STREAM 1

License Optimization

6. License Optimization — Features & Requirements

The License Optimization stream delivers AI-driven analysis of SAP user licenses, role assignments, authorization objects, and transaction usage. It enables Lotte Chemical to identify optimization opportunities, right-size license consumption, and reduce recurring SAP costs.

6.1 In-Scope Features (Statement of Work)

LO-01 User Information Analysis

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI automatically ingests and analyzes SAP user master data including user types, validity dates, lock status, license assignments, and system usage patterns. The platform correlates this with authorization and activity data to build a unified user profile.

Key Data Points

- User Name, User Type, Validity Date (From / To)
- Status
- License Type Assignment
- Last Login Date and Login Frequency

Expected Behavior

- Automated ingestion without manual data extraction from SAP
- Single consolidated user profile view per user
- Filtering and sorting by user type, license, status, and validity
- Downloadable user data in Excel format
- Consistent data across Home Page and Landing Page views

Business Value

Single source of truth for SAP user details. Enables Lotte Chemical to quickly identify inactive or incorrectly classified users, reducing compliance risk.

LO-02 License Analysis & Classification

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI evaluates assigned SAP licenses against actual system usage by mapping user activities, transaction executions, and authorization footprints. The AI engine identifies license mismatches, over-licensed users, and under-utilized license categories.

Key Data Points

- Current License Type (Professional / Functional / Productivity / Self-Service)
- Actual Usage Footprint (transactions executed, objects used)

- License Match/Mismatch Indicator
- Recommended License Type based on usage

Expected Behavior

- AI-driven license match/mismatch detection
- Clear visualization of over-licensed vs. correctly licensed users
- Summary KPI cards showing total licenses by type on Home Page
- Drill-down to individual user license detail on Landing Page
- Export functionality for license analysis data

Business Value

Enables significant SAP license cost optimization by highlighting opportunities to downgrade or reclassify licenses based on real usage.

LO-03 Usage Statistics

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI analyzes historical usage data including login frequency, transaction execution counts, and activity recency to distinguish active users from dormant or infrequently used accounts. Trend detection is automated across time periods.

Key Data Points

- Login Count (Last 30 / 60 / 90 days)
- Transaction Execution Count
- Last Activity Date
- Active / Dormant / Inactive Classification
- Usage Trend (Trending Up / Flat / Declining)

Expected Behavior

- Automated dormancy and inactivity detection
- Time-period based filtering (30/60/90/180/365 days)
- Exportable usage summary report

Business Value

Improves governance by identifying users who no longer require access, supporting proactive cleanup of unused accounts.

LO-04 Assigned Roles Analysis

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI analyzes all roles assigned to users — including composite and single roles — and evaluates redundancy, overlap, and relevance based on actual usage and authorization depth.

Key Data Points

- Role Name
- Role Type (Single / Composite / Derived)
- Assignment Source (Direct / Indirect)
- Usage Status (Used / Unused / Partially Used)
- Redundant Roles (Different Roles with same Auth Object/ Same Roles)

Expected Behavior

- Full role list per user with usage annotation
- Composite role expansion to constituent single roles
- Redundancy detection across user role assignments
- Role assignment filterable by usage status
- Export role assignment data per user or across all users

Business Value

Supports role rationalization and standardization, reducing role sprawl and making the authorization landscape easier to manage and migrate.

LO-05 Authorization Objects Overview

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Consolidated view of all authorization objects assigned to users, including frequency of use. Eliminates need for manual SUIM analysis.

Key Data Points

- Authorization Object Code & Description
- Assigned via Role vs. Directly Assigned
- Usage Count
- Last Used Date

Expected Behavior

- Consolidated authorization object list per user
- Distinction between role-based and directly assigned objects
- Usage frequency and recency (when it was last used) data
- Filterable and sortable by object, usage, and assignment type

Business Value

Improves authorization design transparency, helping IT and security teams eliminate obsolete or excessive authorizations.

LO-06 Assigned Transactions Analysis

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI maps transaction codes (T-Codes) assigned via roles and directly to users, correlating them with execution history to identify unused or rarely used transactions.

Key Data Points

- T-Code
- Assignment Source (Role / Direct)
- Execution Count
- Last Execution Date
- Used / Unused Classification

Expected Behavior

- Full T-Code list per user with execution data
- Clear Used vs. Unused classification
- Cross-reference to assigning roles
- Export T-Code analysis data

Business Value

Allows removal of unnecessary transactional access, lowering risk and supporting least-privilege principles.

LO-07 Directly Assigned Authorization Objects

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Identifies authorization objects directly assigned to users (outside role-based assignments), which are often overlooked in traditional audits.

Key Data Points

- Object Code & Description
- Field Values
- Assignment Date
- Usage Status

Expected Behavior

- Isolated view of direct-assignment-only objects
- Usage trace per directly assigned object
- Highlight objects bypassing role governance
- Export capability

Business Value

Eliminates undocumented and non-standard access paths, strengthening governance and ensuring consistent role-based security practices.

LO-08 Used & Unused Authorization Objects

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI tracks which authorization objects are actually used during system execution, distinguishing them from merely assigned permissions. Also surfaces Unused Directly Assigned Authorization Objects that are assigned to users but never triggered at runtime.

Key Data Points

- Object Code
- Used / Unused Flag
- Execution Count
- Last Used Date
- Assignment Source

Expected Behavior

- Side-by-side view of assigned vs. used objects
- Drill-down from used objects to specific transactions
- Export both used and unused object lists

Business Value

Supports precise authorization cleanup by retaining only what is functionally required, reducing complexity.

Note — Dependency

Authorization Object Field Value based usage analysis requires Lotte Chemical to have activated the required trace mechanisms in the SAP System prior to each Assessment run.

LO-09 License Breakdown

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Granular breakdown of license types across all users, mapping each license category to actual usage behavior and access patterns.

Key Data Points

- License Type
- User Count by License Type
- Usage Match Rate by License Type

Expected Behavior

- Summary breakdown chart/table on Home Page dashboard
- Drill-down to individual users per license type
- Trend view across assessment runs
- Export breakdown data

Business Value

Gives leadership clear visibility into where SAP license spend is concentrated, enabling informed reclassification decisions.

LO-10 Top Transactions & Unused Transactions

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI analyzes execution logs to identify the most frequently used SAP transactions and those that are assigned but never executed.

Key Data Points

- T-Code
- Execution Count
- License Type Contribution

Expected Behavior

- Top N transactions by usage across all users
- Unused transactions list per user and globally
- Filterable by department, user type, license type
- Export both views

Business Value

Distinguishes core business transactions from peripheral usage; enables tighter access control and accurate license entitlement alignment.

LO-11 Complete Role Utilization Analysis

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI evaluates role utilization across all assigned roles, measuring actual usage of transactions and authorization objects within each role. Also extracts structural data from role, authorization object, field, and value levels.

Key Data Points

- Role Name
- Assigned User Names
- Used / Unused Classification
- Object Count (Total / Used / Unused)
- Authorization Object, Field, Value Structure

Expected Behavior

- Role utilization summary table
- Drill-down from role to contributing users and objects
- Export role utilization data with full structural breakdown
- Filterable by utilization level and role type

Business Value

Supports role consolidation and standardization; critical for SAP scalability and S/4HANA readiness.

LO-12 Role Activity & Authorization Analysis

✓ **IN SCOPE (SoW)** Covered under Statement of Work

Description

KTern.AI analyzes SAP roles at a granular authorization-object and activity level by decoding role definitions, authorization objects, fields, and access types (Display, Post, Change, etc.).

Key Data Points

- Activity Type (01-Create, 02-Change, 03-Display, etc.)
- Authorization Object per Activity

Expected Behavior

- Activity-level breakdown per role
- Export role activity matrix

Business Value

Enables stronger governance; ensures minimum required access; helps reduce compliance risks and prevent unauthorized postings.

6.2 Enhancements (Enhancement Tickets — Governed by SLA)

The following enhancements are not covered in the signed SoW. Each will be raised as a separate enhancement ticket and delivered per the KTern.AI SLA. Priority classification and delivery timelines will be agreed with the Lotte Chemical project team upon ticket submission.

LO-ENH-01 Target License Classification Column

⚠ **Enhancement Ticket Required** Requires separate enhancement ticket per KTern.AI SLA

Ticket Summary

Introduce a new column "Target License Classification" after the Email column in the user table. This column represents the highest privileged license type assigned to each user.

Classification Logic

- Hierarchy: Productivity < Functional < Professional
- If a user has multiple license types → assign the highest applicable
- Professional always takes precedence → classify as Professional
- Functional + Productivity → classify as Functional
- Only Productivity → classify as Productivity

Expected Behavior

- Column appears after Email column in user table
- Logic computed at backend level for consistency
- Available in UI, included in export/download
- Consistent across Home Page and Landing Page views
- Filterable and sortable by Target License Classification

Business Impact

- Provides clear, standardized single-value license classification per user
- Enables accurate license optimization and cost reduction decisions
- Reduces manual consolidation effort — improves operational efficiency
- Improves reporting clarity for stakeholders — better governance and audit readiness

Priority Suggestion

P2 — High (directly impacts License Optimization accuracy)

LO-ENH-02 User-Based License Consumption Card (Home Page)

⚠ **Enhancement Ticket Required** Requires separate enhancement ticket per KTern.AI SLA

Ticket Summary

Add a "User-Based License Consumption" card to the License Optimization Dashboard Home Page. This card provides a tree-view breakdown of license consumption per user.

Required Columns / Data Points

- User Name
- Number of Professional Licenses
- Number of Functional Licenses
- Number of Productive / Core Licenses
- Target License Classification (per LO-ENH-01 logic)

Expected Behavior

- Tree-view structure with expand/collapse at user level
- Expand All and Collapse All controls
- Auto drill-down to contributing roles or authorization objects
- Highlight final Target License Classification per user
- Filter capability (by user, license type, classification)
- Download/export option (Excel)
- Consistent with Landing Page data

Business Impact

- Single consolidated user-level view of license consumption
- Faster and more accurate license optimization decisions
- Reduced manual effort and improved transparency
- Improves audit readiness and stakeholder reporting

Reference

See Image Reference 1 (uploaded reference screenshots — column structure to follow LO-ENH-01 logic).

Priority Suggestion

P2 — High

LO-ENH-03 Role-Based License Classification Card (Home Page)

⚠ Enhancement Ticket Required Requires separate enhancement ticket per KTern.AI SLA

Ticket Summary

Add a "Role-Based License Classification" card to the License Optimization Dashboard Home Page. This card shows the Target License Classification per role.

Required Columns / Data Points

- Role Name

- Role Description
- Target License Classification

Expected Behavior

- Role list with description and classification
- Indicate target license classification per role clearly
- Filter capability (by role name, classification)
- Download/export option (Excel)
- Consistent with Landing Page data

Business Impact

- Clear understanding of role-based license classification
- Supports role optimization and cleanup efforts
- Improves governance and compliance visibility
- Reduces manual analysis effort

Reference

See Image Reference 2 (uploaded reference screenshots).

Fallback Option

If new cards are not immediately feasible: Add Target License Classification column and license count columns (in tree view) to the existing table. Allow users to hide/show columns. Ensure only selected columns / filtered data are downloaded.

Priority Suggestion

P2 — High

LO-ENH-04 AI-Driven Recommendations on License Optimization Dashboard & Export

⚠ Enhancement Ticket Required Requires separate enhancement ticket per KTern.AI SLA

Ticket Summary

Surface license optimization recommendations on both the Dashboard Home Page and Landing Page, and include them in the exported Excel/PDF report. At a minimum, basic recommendations must be visible.

Recommendation Types

- Object-Level Cleaning (Focus: Professional License Objects)
- Role-Based Cleaning (Focus: Usage Efficiency)
- Authorization-Based Role Optimization (Focus: Redundant Role Cleanup)

Recommendation 1: Object-Level Cleaning

- Identify all Professional license authorization objects
- Trace actual object usage per user
- Classify users: Professional Required vs. Functional/Productive Sufficient
- Step 1 — Object Traceability: Identify all roles containing the auth object (irrespective of used/unused)
- Step 2 — Role Assignment: For each role containing the object, state number of users assigned and list user names
- Step 3 — Usage & License Segregation: Identify users who actually use the object; classify into High Privilege (Professional required) vs. Lower Privilege (Functional/Productive sufficient)
- Step 4 — Role Optimization: Create a new high-privilege role for Professional users; remove high-privilege objects from old role; retain only required objects for remaining users in the old role

Recommendation 2: Role-Based Cleaning

- Identify all roles assigned to users
- Step 1 — Role Assignment Analysis: Identify all users assigned to the role
- Step 2 — Usage Validation: Identify users who did not use any objects in the role or have very low usage
- Step 3 — Pattern Identification: If majority unused → role is cleanup candidate; if partially used → role requires optimization (object cleanup)
- Step 4 — User-Level Cleanup: Recommend removal of unused role assignments for users not using the role

Recommendation 3: Authorization-Based Role Optimization

- Identify authorization objects present across multiple roles
- Detect object overlap across roles
- Logic: If object exists in multiple roles → do not mark all roles as used by that object alone
- If role has only redundant objects OR only unused objects → recommend role removal
- If role has at least one unique used object → retain role
- Example: User has Role A (O1 + unused objects) and Role B (O1 only). O1 is used → Remove Role A as O1 is covered by Role B

Dashboard Behavior

- Recommendations panel/section on Home Page and Landing Page
- Each recommendation includes: User/Role, Recommendation Type, Recommended Action, Justification
- Filterable by recommendation type and severity

- Recommendations included in Excel/PDF export

Business Impact

- Data without recommendation is not actionable — this closes the analysis-to-action gap
- Faster decisions and higher adoption of KTern.AI platform
- Better trust in system output for stakeholders and auditors

Priority Suggestion

P3 — Medium (foundational to License Optimization value; recommend prioritizing in next monthly release)

Note: Any run with the result report having issues shouldn't be counted part of the 12 run.

STREAM 2

Segregation of Duties (SoD) Analysis

7. SoD Analysis — Features & Requirements

The SoD Analysis stream delivers AI-powered detection and assessment of Segregation of Duties conflicts, critical access violations, and compliance gaps across Lotte Chemical's SAP authorization landscape. It enables proactive compliance governance aligned to SAP GRC best practices.

Important Note:

- SoD compliance analysis will be performed based on custom exception scenarios defined as part of Lotte Chemical's organization.
- Custom Exception Scenarios must be shared with the KTern.AI team at least 10 business days prior to each assessment run.

7.1 In-Scope Features (Statement of Work)

SOD-01 Executive Summary Dashboard

✓ **IN SCOPE (SoW)** Covered under Statement of Work

Description

KTern.AI delivers an automated, data-driven SoD compliance assessment summary. The executive summary presents AI-powered risk classification, critical violation highlights, and a business-consumable compliance overview aligned to SAP GRC best practices.

Key Data Points

- Total Users Analyzed
- Total SoD Violations Detected
- Violations by Severity (Critical / High / Medium / Low)
- Top Risk Categories
- Overall Compliance Score

Expected Behavior

- Single-screen executive summary visible on SoD Dashboard Home Page
- KPI cards for total violations, severity breakdown, and compliance score
- Visual charts (pie/bar) for risk distribution

- Drill-down to detailed violation list
- Export executive summary as PDF/Excel

Business Value

Provides leadership with immediate enterprise-wide access risk visibility, enabling faster governance decisions.

SOD-02 Overall Compliance Assessment

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI computes an overall compliance posture by correlating user access, role design, and transactional behavior against predefined SoD risk matrices, benchmarked against SAP and industry standards.

Key Data Points

- Compliance Maturity Score
- Benchmark vs. SAP GRC Standard
- Compliance Trend (Improvement / Deterioration across runs)
- Systemic Weakness Identification

Expected Behavior

- Compliance posture scorecard with trend
- Benchmark comparison panel
- Historical run comparison across assessment runs
- Export compliance assessment report

Business Value

Enables Lotte to clearly understand compliance maturity and define a roadmap toward audit readiness.

SOD-03 Critical Findings Summary

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Automatically consolidates all high-risk and critical violations across users, roles, and processes into a unified risk summary. AI-driven severity scoring ensures accurate prioritization.

Key Data Points

- Violation ID
- Violation Description
- Affected Users
- Severity Score
- Process Area (Finance / Procurement / OTC / HR / IT)
- Recommended Action

Expected Behavior

- Consolidated critical findings list on Home Page
- Severity-based sorting and filtering
- Drill-down to individual violation detail
- Export critical findings list

Business Value

Reduces dependency on manual audit interpretation; enables focused remediation on the highest-impact risks.

SOD-04 Immediate Actions Required (24–48 Hours)

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

KTern.AI identifies violations requiring urgent remediation and provides specific actionable steps such as access revocation, role redesign, or emergency access governance.

Key Data Points

- Violation Description
- Urgency Level
- Affected User(s)
- Recommended Immediate Action
- Estimated Business Risk if Unaddressed

Expected Behavior

- Distinct "Immediate Actions" panel on SoD Dashboard
- Clear call-to-action per violation
- Assignable to responsible team/user
- Track status of immediate action completion
- Export immediate actions list

Business Value

Empowers Lotte to respond quickly to high-risk audit issues, protecting financial integrity.

SOD-05 Compliance Impact Assessment

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Maps identified SoD violations to their potential financial, operational, and regulatory impact using rule-based and AI-assisted analysis.

Key Data Points

- Impact Category (Financial / Operational / Regulatory)
- Estimated Exposure Level
- Linked Process Areas
- Regulatory Framework Reference (SOX, GDPR, etc.)

Expected Behavior

- Impact matrix view linking violations to business impact
- Risk quantification panel
- Regulatory framework mapping
- Export impact assessment

Business Value

Helps leadership quantify compliance risk exposure and justify remediation investments.

SOD-06 Critical Violations — Category 1: Super Administrators

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Detects users with unrestricted or near-unrestricted access across SAP systems, highlighting concentration-of-power risks and non-compliant access patterns.

Key Data Points

- User Name / ID
- Super Admin Indicator
- Roles Contributing to Super Admin Status
- Recommendation (Revoke / Redesign / Monitor)

Expected Behavior

- Dedicated Super Administrator violations panel
- User list with contributing role details
- Severity flagging and recommended actions
- Export super admin violation list

Business Value

Reduces risk of unauthorized system changes, fraud, and audit failure by enforcing least-privilege access.

SOD-07 Critical Violations — Category 2: Dual Process Control

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Identifies users controlling multiple end-to-end business processes through cross-role and transaction analysis based on SAP standard dual process control violations.

Key Data Points

- User Name
- Process 1 (e.g., Create Vendor)
- Process 2 (e.g., Approve Payment)
- Conflicting Transactions
- Severity (Critical / High)

Expected Behavior

- Dual process conflict matrix view
- User-level violation detail with conflicting transactions
- Process area drill-down
- Export dual process violation list

Business Value

Mitigates financial fraud risk and ensures integrity across critical revenue and procurement cycles.

SOD-08 Critical Violations — Category 3: Emergency Access

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Analyzes firefighter and emergency access usage patterns, detecting excessive access, prolonged assignments, and missing approvals.

Key Data Points

- User Name
- Firefighter ID / Emergency Role
- Assignment Duration
- Usage Frequency
- Approval Status
- Anomaly Flag

Expected Behavior

- Emergency access usage timeline per user
- Approval gap detection
- Prolonged assignment alerting
- Export emergency access report

Business Value

Ensures emergency access is used only for justified scenarios, reducing audit exposure and preventing misuse of privileged access.

SOD-09 Critical Violations — Category 4: Complete OTC Control

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Flags users with end-to-end Order-to-Cash control, combining transactional authority and financial posting privileges across the OTC process cycle.

Key Data Points

- User Name
- OTC Steps Controlled (Order Entry / Delivery / Billing / Collection)
- Conflicting Transactions
- Financial Impact Exposure

Expected Behavior

- OTC control violation panel
- Process step coverage visualization
- User drill-down with conflicting transaction list
- Export OTC control violations

Business Value

Protects revenue integrity and prevents manipulation of sales, billing, and collections processes.

SOD-10 High-Risk Violations — Service Accounts

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Identifies service and technical accounts with excessive or unmanaged privileges, including background job and integration users.

Key Data Points

- Account Name / ID
- Account Type (Service / Background / Integration)
- Privilege Level
- Last Activity
- Risk Classification

Expected Behavior

- Service account violation list
- Privilege level indicator
- Inactivity and excessive access flags
- Export service account violation report

Business Value

Reduces cyber and operational risk by tightening control over non-human access, a common blind spot in audits.

SOD-11 Remediation & Governance Recommendations

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Provides AI-backed remediation recommendations including role redesign, access removal, mitigating controls, and governance policy suggestions.

Key Data Points

- Violation ID
- Recommendation Type (Role Redesign / Access Removal / Mitigating Control / Policy)
- Action Steps
- Priority
- Assignee Suggestion

Expected Behavior

- Remediation recommendations panel on SoD Dashboard
- Actionable steps per recommendation
- Assignable to responsible team
- Track recommendation status (Open / In Progress / Resolved)
- Export remediation plan

Business Value

Accelerates remediation cycles and reduces reliance on external auditors; enables sustainable compliance operations.

SOD-12 Continuous Compliance Enablement

✓ IN SCOPE (SoW) Covered under Statement of Work

Description

Supports ongoing monitoring and re-assessment through periodic SoD scans and live dashboards, enabling continuous compliance monitoring.

Key Data Points

- Scan Run Date
- Violations Trend (New / Resolved / Persisting)
- Compliance Score Trend

- Assessment Run History

Expected Behavior

- Multiple assessment run support (up to 90 runs/year as per SoW)
- Trend comparison across runs
- Compliance improvement tracking
- Historical run archive with drill-down
- Automated alerts for new critical violations (if configured)

Business Value

Moves Lotte from point-in-time audits to proactive, continuous compliance — reducing future audit cost and risk.

7.2 Enhancements — SoD Analysis

No enhancement tickets have been raised for the SoD Analysis stream at this time. Any enhancements identified during the engagement will be submitted as enhancement tickets per the SLA process described in Section 5.

8. Non-Functional Requirements

8.1 Performance

Requirement	Specification
Dashboard Load Time	Home Page dashboard must load within 5 seconds for up to 10,000 users
Assessment Run Duration	Full License Optimization run to complete within 4 hours for Lotte Chemical data volumes
Export Generation	Excel / PDF export to generate within 60 seconds for standard data volumes
Concurrent Users	Platform must support minimum 50 concurrent dashboard users without performance degradation
API Response Time	All internal API calls to return within 3 seconds under normal load

8.2 Security & Compliance

Requirement	Specification
Data Encryption	All data at rest and in transit must be encrypted (AES-256 / TLS 1.2+)

Authentication	SSO and multi-factor authentication (MFA) support
Role-Based Access Control	Platform access governed by RBAC with tenant-level isolation
Data Residency	SAP data processed in compliance with Lotte Chemical data residency policies
Audit Logging	All user actions and data access events logged and auditable
SAP Certification	Platform maintains SAP AI Certified Platform status
Architecture	Require information on the architecture part, where and all it goes & access the data

8.3 Availability & SLA

Requirement	Specification
Platform Availability	99.9% uptime SLA (24x7), excluding planned maintenance windows
Planned Maintenance	Advance electronic notice provided before scheduled downtime
Assessment Runs	License Optimization: 12 runs/year; SoD: 90 runs/year (per SoW commercials)
Support Hours	KTern.AI Customer Success support during business hours; critical support 9x5
Data Re-run Policy	Re-run available at no count cost for validated data discrepancy scenarios (per SoW T&C)

9. Requirements Traceability Matrix (RTM)

The following matrix maps each feature to its SoW coverage status, priority, and business owner.

Feature ID	Feature Name	Stream	SoW Coverage	Priority	Business Owner
LO-01	User Information Analysis	License Opt.	In Scope	P1	SAP License Manager

LO-02	License Analysis & Classification	License Opt.	In Scope	P1	SAP License Manager
LO-03	Usage Statistics	License Opt.	In Scope	P1	SAP Basis Team
LO-04	Assigned Roles Analysis	License Opt.	In Scope	P1	SAP Security Team
LO-05	Authorization Objects Overview	License Opt.	In Scope	P1	SAP Security Team
LO-06	Assigned Transactions Analysis	License Opt.	In Scope	P1	SAP Basis Team
LO-07	Directly Assigned Auth Objects	License Opt.	In Scope	P1	SAP Security Team
LO-08	Used & Unused Auth Objects	License Opt.	In Scope	P1	SAP Security Team
LO-09	License Breakdown	License Opt.	In Scope	P1	SAP License Manager
LO-10	Top Transactions & Unused	License Opt.	In Scope	P2	SAP Basis Team
LO-11	Complete Role Utilization Analysis	License Opt.	In Scope	P1	SAP Security Team
LO-12	Role Activity & Auth Analysis	License Opt.	In Scope	P1	SAP Security Team
LO-ENH-01	Target License Classification Column	License Opt.	ENHANCEMENT	P2	SAP License Manager
LO-ENH-02	User-Based License Consumption Card	License Opt.	ENHANCEMENT	P2	SAP License Manager
LO-ENH-03	Role-Based License Classification Card	License Opt.	ENHANCEMENT	P2	SAP License Manager
LO-ENH-04	AI Recommendations on Dashboard	License Opt.	ENHANCEMENT	P3	SAP License Manager
SOD-01	Executive Summary Dashboard	SoD Analysis	In Scope	P1	IT Compliance
SOD-02	Overall Compliance Assessment	SoD Analysis	In Scope	P1	IT Compliance

SOD-03	Critical Findings Summary	SoD Analysis	In Scope	P1	IT Compliance
SOD-04	Immediate Actions (24–48 hrs)	SoD Analysis	In Scope	P1	IT Compliance
SOD-05	Compliance Impact Assessment	SoD Analysis	In Scope	P1	Finance / Risk
SOD-06	Super Administrators Detection	SoD Analysis	In Scope	P1	IT Compliance
SOD-07	Dual Process Control Violations	SoD Analysis	In Scope	P1	Finance / Risk
SOD-08	Emergency Access Violations	SoD Analysis	In Scope	P1	IT Compliance
SOD-09	OTC Control Violations	SoD Analysis	In Scope	P1	Finance / Risk
SOD-10	High-Risk Service Accounts	SoD Analysis	In Scope	P2	SAP Basis Team
SOD-11	Remediation & Governance Recs	SoD Analysis	In Scope	P1	IT Compliance
SOD-12	Continuous Compliance Enablement	SoD Analysis	In Scope	P2	IT Compliance

10. Commercials Summary

Parameter	Details
Solution	KTern.AI — License Optimization & SoD Agents
Contract Duration	3 years (March 4, 2026 – March 4, 2029)
Total Price	USD 150,000 (inclusive of unlimited app user access and customer support)
License Optimization Runs	12 runs per year
SoD Analysis Runs	90 runs per year
Billing Milestone 1	USD 50,000 — March 4, 2026
Billing Milestone 2	USD 50,000 — March 4, 2027
Billing Milestone 3	USD 50,000 — March 4, 2028
Enhancement Tickets	Governed by KTern.AI SLA (https://ktern.com/trust-center/sla)

11. Glossary

Term	Definition
Authorization Object	An SAP security element that controls access to business functions, fields, and activities within the system.
Enhancement Ticket	A formal request for a feature or change that is outside the signed SoW, processed under the KTern.AI SLA.
Firefighter Access	Emergency privileged access in SAP, typically managed via SAP GRC firefighter IDs for break-glass scenarios.
License Classification	The categorization of an SAP user's license type (Professional / Functional / Productivity) based on their access and usage patterns.
OTC	Order-to-Cash — the end-to-end business process from customer order creation through billing and cash collection.
PRD	Product Requirements Document — this document.
Role Utilization	The degree to which authorization objects and transactions within an assigned role are actively used by the assigned user.
SLA	Service Level Agreement — the KTern.AI policy governing support response times and enhancement ticket delivery.
SoD	Segregation of Duties — the principle of separating conflicting business process steps across different users to reduce fraud and error risk.
SoW	Statement of Work — the signed contractual document between KTern.AI and Lotte Chemical defining the scope of delivery.
T-Code	Transaction Code — an SAP shortcut that executes a specific SAP function or program.
USMM / LAW	SAP License Administration Workbench and User System Measurement — SAP tools for license measurement and compliance.